Practice workbook · Sound sampling: analogue to digital

How analogue sound becomes digital: the sampling process, reading/plotting a sampled waveform and its axes, sampling frequency and the effect of changing it.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: CSC-T3-13

1. Question 1 · part c.i · 3 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part c.i only.

1 Computers use binary to represent and store data.

(a) Binary and hexadecimal notation can be used to represent numbers.

(i) Convert the denary number 77 to 8-bit binary.

(2)

(ii) Convert the denary number –126 to 8-bit binary using two's complement.

(2)

OLevels.COM — SCHOOL —

(iii) Convert the binary pattern 11000110 to hexadecimal.

(2)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

1. Question 1 · part c.i · 3 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part c.i only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(b) Images are stored as binary data.

(i) A bitmap image is 400 pixels wide by 200 pixels high.

It has a colour depth of 12 bits.

Construct an expression to show how the image size, in bytes, is calculated.

Do **not** calculate the answer.

(2)

(ii) **Figure 1** is an image of an icon, set out in an 8 × 8 pixel grid.

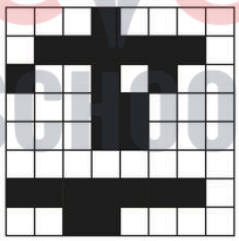


Figure 1

A run-length encoding compression algorithm uses:

- W for white pixels
- B for black pixels.

Give the result of compressing the first 4 lines of the icon using the algorithm.

(3)

Line 1 .....

Line 2 .....

Line 3 .....

Line 4 .....

OLevels.COM SCHOOL

1. Question 1 · part c.i · 3 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part c.i only.

(iii) Run-length encoding is a type of lossless compression.

State what is meant by the term **lossless** in this context.

(1)

(c) Raoul wants to record a song and store it as a digital sound file.

Figure 2 shows an overview of the process of recording a song and storing it as a digital sound file.

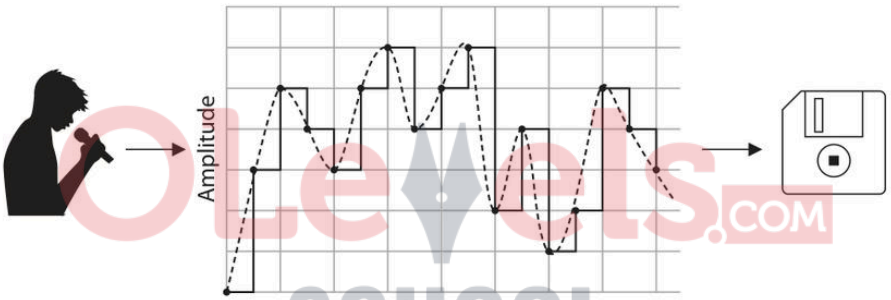


Figure 2

(i) Describe the steps taken to convert the analogue sound to a digital sound file.

(3)

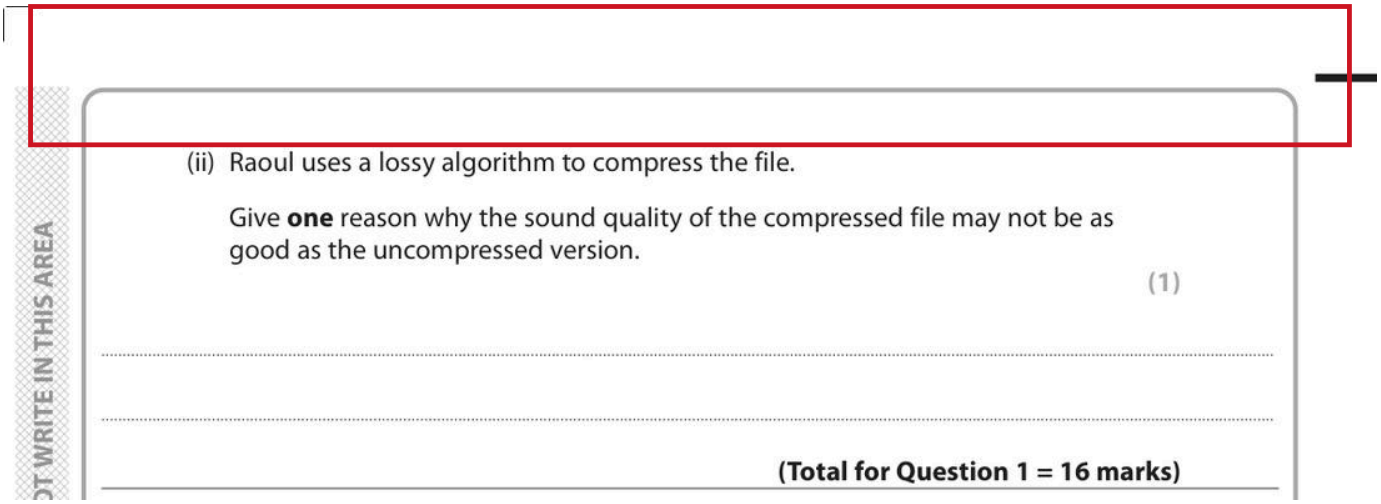
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 6 9 2 9 7 A 0 4 2 0

1. Question 1 · part c.i · 3 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part c.i only.



DO NOT WRITE IN THIS AREA

(ii) Raoul uses a lossy algorithm to compress the file.

Give **one** reason why the sound quality of the compressed file may not be as good as the uncompressed version.

(1)

.....

.....

(Total for Question 1 = 16 marks)

Working and answer

OLevels.COM — SCHOOL —

2. Question 4 · part a.i · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part a.i only.

4 Ibrahim works in the music industry.

(a) He is learning how sound is represented in binary.

Figure 3 illustrates this process.

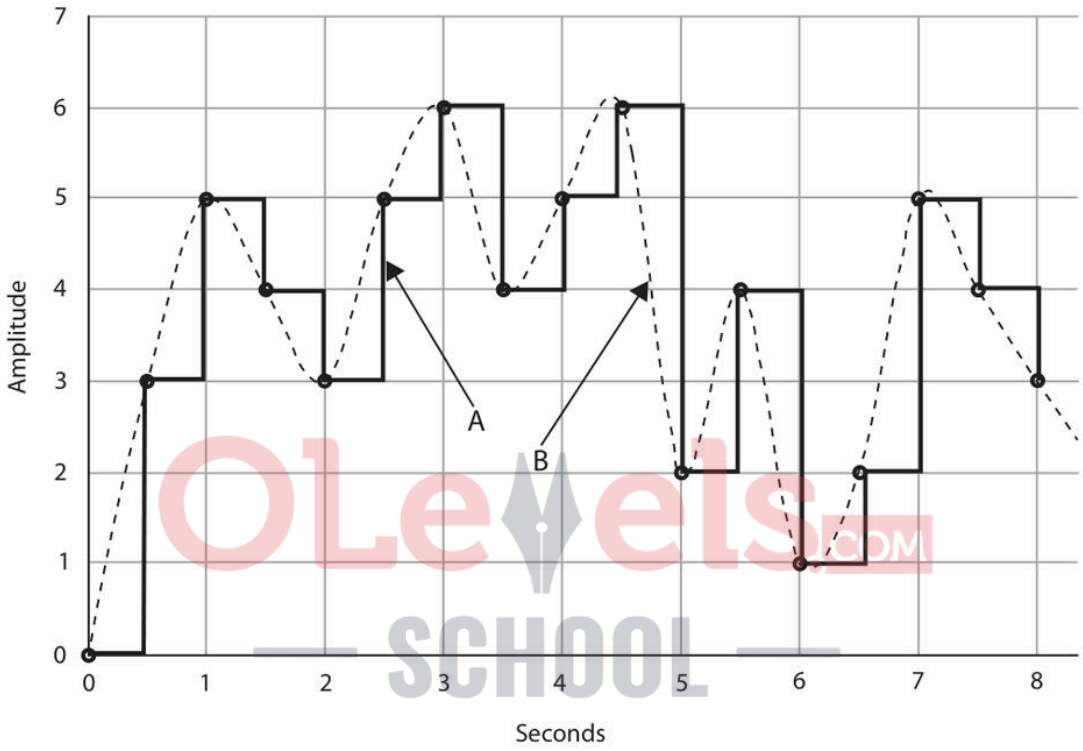


Figure 3

(i) State what the solid line labelled **A** represents.

(1)

(ii) State what the dashed line labelled **B** represents.

(1)

(iii) Give the sampling frequency, including the correct units.

(1)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

P 6 1 8 8 4 R R A 0 1 0 2 0

2. Question 4 · part a.i · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 4 · part a.ii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part a.ii only.

4 Ibrahim works in the music industry.

(a) He is learning how sound is represented in binary.

Figure 3 illustrates this process.

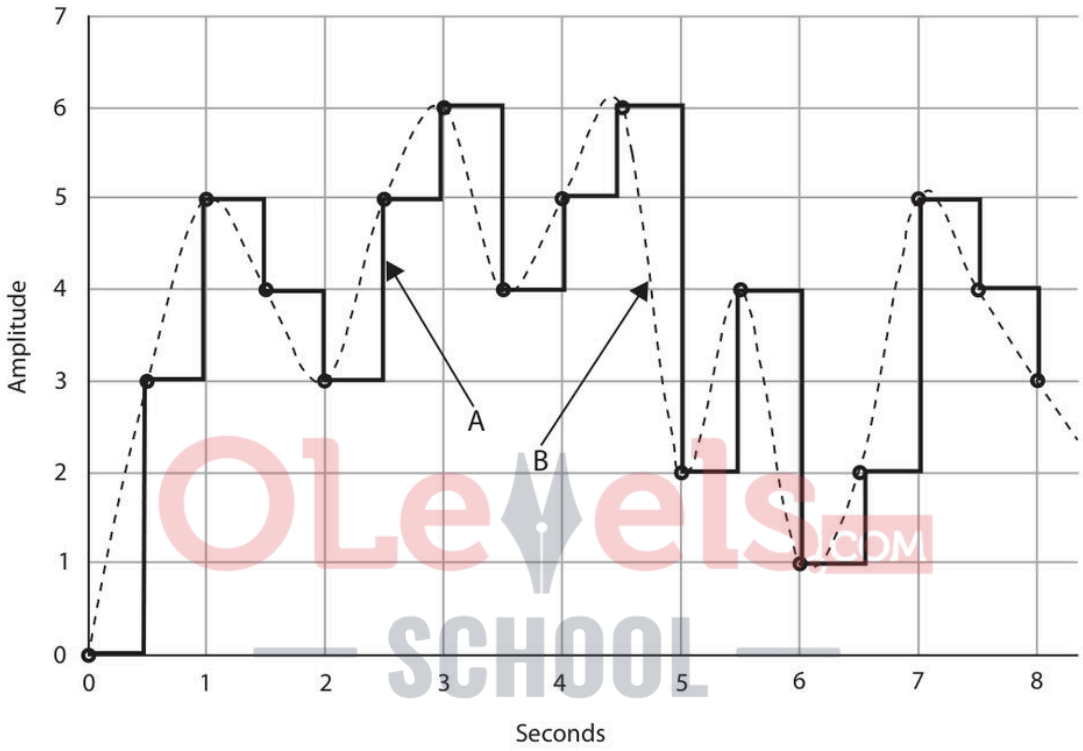


Figure 3

(i) State what the solid line labelled **A** represents.

(1)

(ii) State what the dashed line labelled **B** represents.

(1)

(iii) Give the sampling frequency, including the correct units.

(1)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

P 6 1 8 8 4 R R A 0 1 0 2 0

3. Question 4 · part a.ii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 4 · part a.iii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part a.iii only.

4 Ibrahim works in the music industry.

(a) He is learning how sound is represented in binary.

Figure 3 illustrates this process.

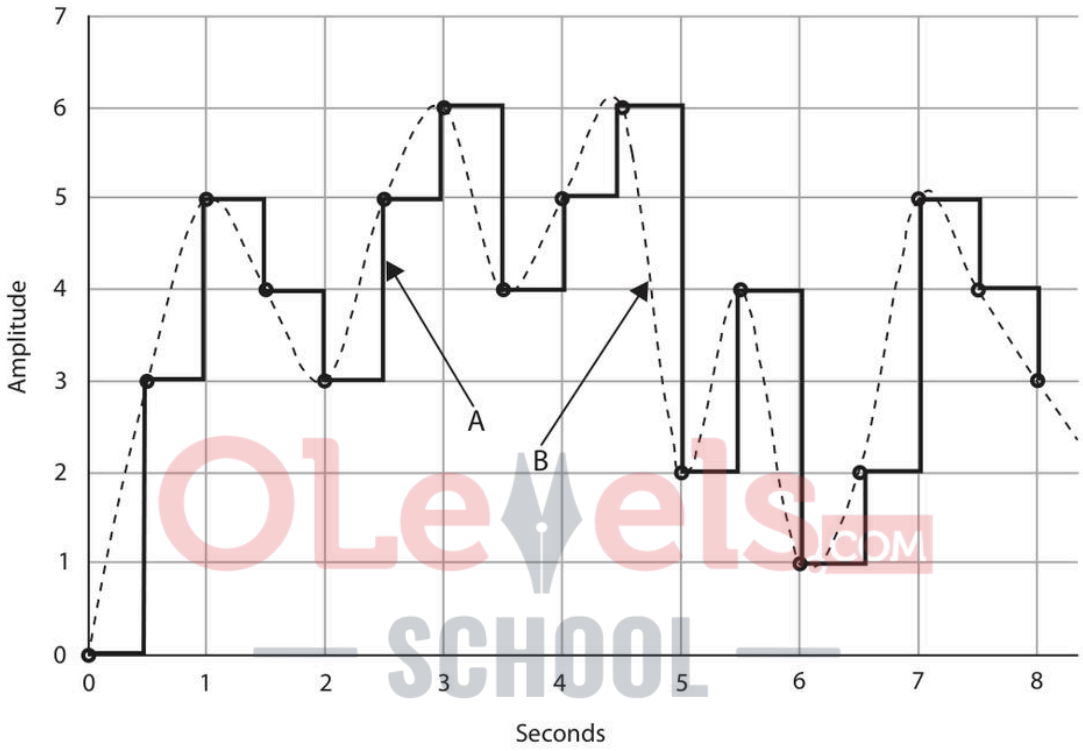


Figure 3

(i) State what the solid line labelled **A** represents.

(1)

(ii) State what the dashed line labelled **B** represents.

(1)

(iii) Give the sampling frequency, including the correct units.

(1)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

P 6 1 8 8 4 R R A 0 1 0 2 0

4. Question 4 · part a.iii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part a.iii only.

(iv) Give **one** impact of increasing the sampling frequency.

(1)

(v) The bit-depth measures the resolution of the sound sample.

State the minimum bit-depth needed to store the sound sampled in **Figure 3**.

Justify your answer.

(2)

Bit-depth

Justification

(vi) Give the value in binary of the audio sample at the 4th second.

(1)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

4. Question 4 · part a.iii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 4 · part a.iv · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part a.iv only.

4 Ibrahim works in the music industry.

(a) He is learning how sound is represented in binary.

Figure 3 illustrates this process.

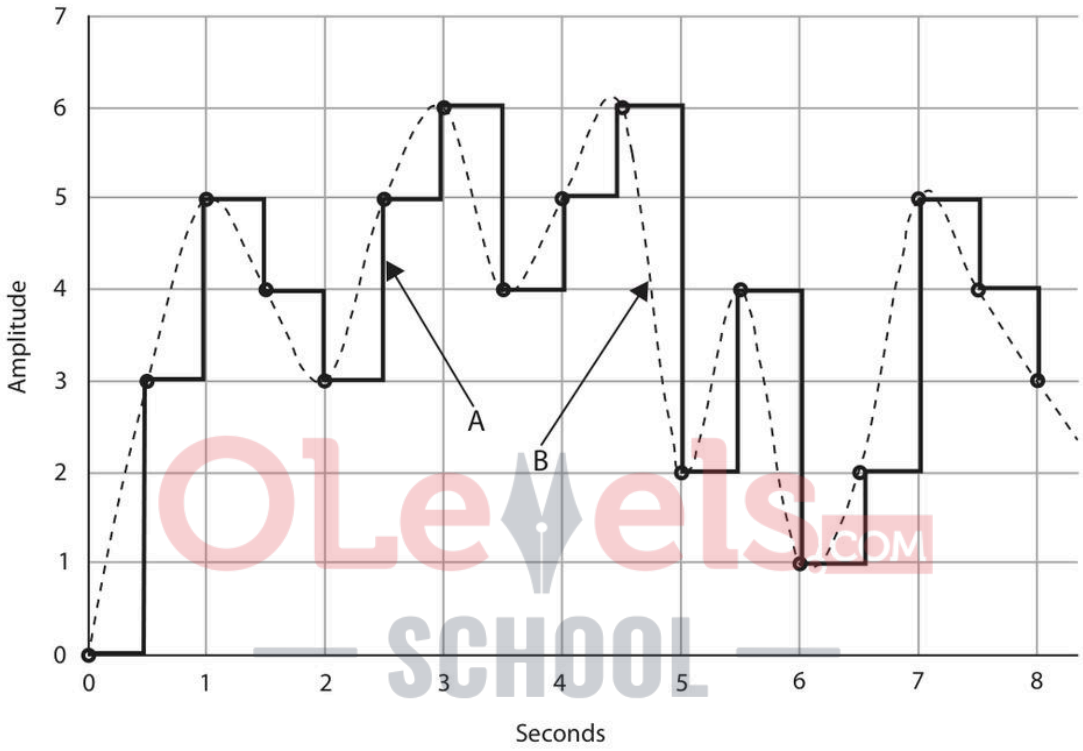


Figure 3

(i) State what the solid line labelled **A** represents.

(1)

(ii) State what the dashed line labelled **B** represents.

(1)

(iii) Give the sampling frequency, including the correct units.

(1)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

10

5. Question 4 · part a.iv · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part a.iv only.

(iv) Give **one** impact of increasing the sampling frequency.

(1)

(v) The bit-depth measures the resolution of the sound sample.

State the minimum bit-depth needed to store the sound sampled in **Figure 3**.

Justify your answer.

(2)

Bit-depth

Justification

(vi) Give the value in binary of the audio sample at the 4th second.

(1)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

5. Question 4 · part a.iv · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 3 · part a.i · 3 marks

2022 JUN · Paper 01

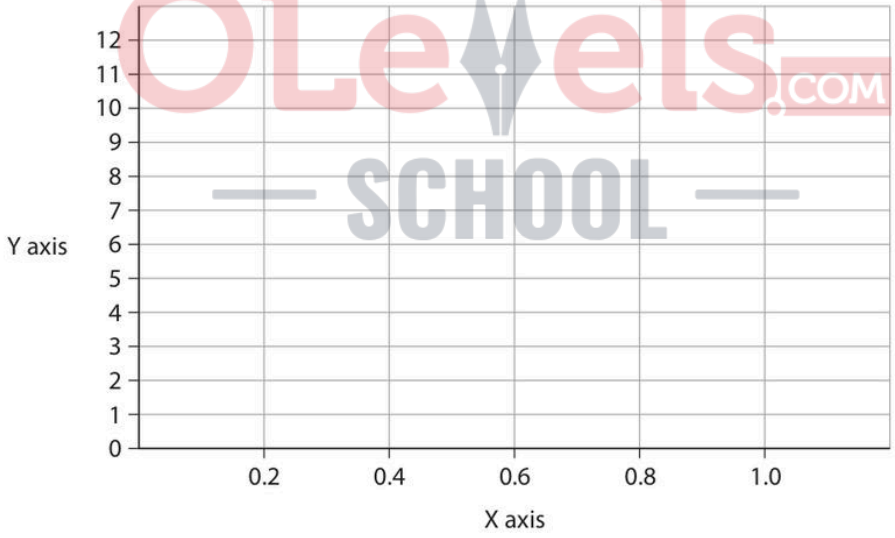
Further practice · Attempt part a.i only.

3 Alyssa is a music producer.

(a) **Figure 1** shows the denary values of five samples of an analogue sound using a sample interval of 0.2 seconds.

Sample number	Denary value
1	1
2	10
3	12
4	5
5	3

Figure 1



(i) Complete this graph using the sample information from **Figure 1** to show the digital sound wave.

(3)

(ii) Give a suitable label for the X axis.

(1)

(iii) Give a suitable label for the Y axis.

(1)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

6. Question 3 · part a.i · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 3 · part a.ii · 1 mark

2022 JUN · Paper 01

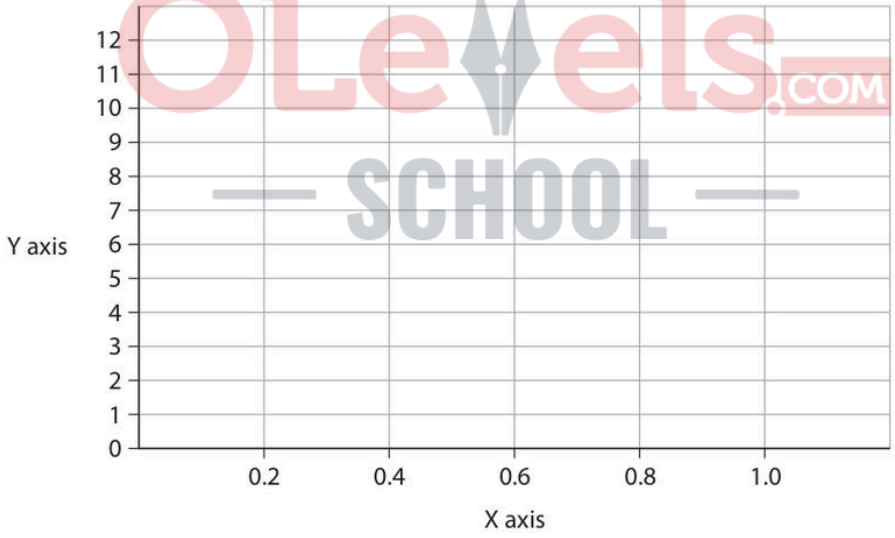
Further practice · Attempt part a.ii only.

3 Alyssa is a music producer.

(a) **Figure 1** shows the denary values of five samples of an analogue sound using a sample interval of 0.2 seconds.

Sample number	Denary value
1	1
2	10
3	12
4	5
5	3

Figure 1



(i) Complete this graph using the sample information from **Figure 1** to show the digital sound wave.

(3)

(ii) Give a suitable label for the X axis.

(1)

.....

(iii) Give a suitable label for the Y axis.

(1)

.....

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

P 7 2 5 3 8 A 0 8 2 0

7. Question 3 · part a.ii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 3 · part a.iii · 1 mark

2022 JUN · Paper 01

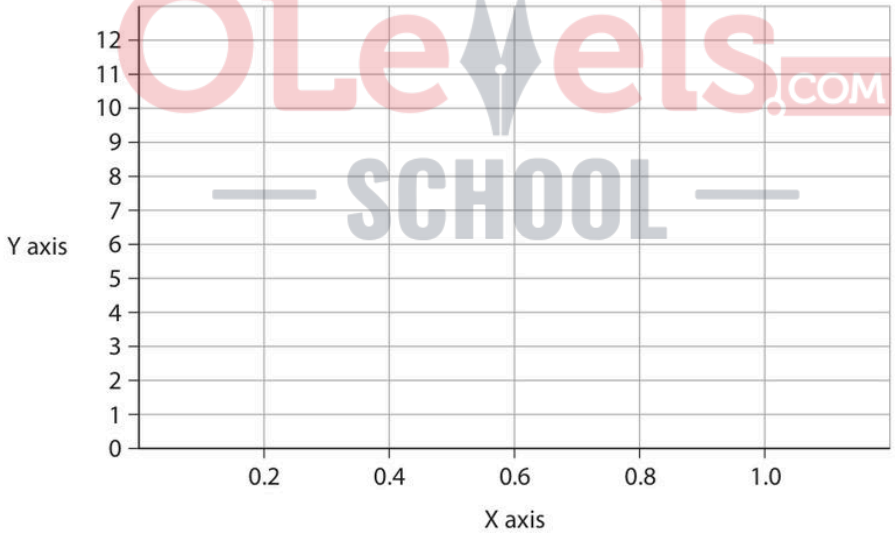
Further practice · Attempt part a.iii only.

3 Alyssa is a music producer.

(a) **Figure 1** shows the denary values of five samples of an analogue sound using a sample interval of 0.2 seconds.

Sample number	Denary value
1	1
2	10
3	12
4	5
5	3

Figure 1



(i) Complete this graph using the sample information from **Figure 1** to show the digital sound wave.

(3)

(ii) Give a suitable label for the X axis.

(1)

(iii) Give a suitable label for the Y axis.

(1)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

8

P 7 2 5 3 8 A 0 8 2 0

8. Question 3 · part a.iii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part a.iii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(b) Alyssa uploads music files to her cloud storage.

(i) She compresses the files before she uploads them using a lossless algorithm.

Give **one** disadvantage of using a lossless rather than a lossy algorithm for this purpose.

(1)

.....

.....

(ii) Explain **one** benefit to Alyssa of storing her music files in the cloud.

(2)

.....

.....

.....

.....

OLevels.COM SCHOOL

(iii) Give **one** possible security issue associated with storing music files in the cloud.

(1)

.....

.....

(iv) One of Alyssa's music files is stored at https://www.cloudisfab.com/re12/ru2.mp3

Complete the table by adding a description of each URL component.

(4)

URL component	Description
https	
www.cloudisfab.com	
re12	
ru2.mp3	

(Total for Question 3 = 13 marks)

8. Question 3 · part a.iii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL